

Import Data Using Transform Maps

Project Description:

This project shows how spreadsheet data can be imported into ServiceNow quickly and correctly using Import Sets and Transform Maps. Instead of typing rows into a form one by one, the user uploads an Excel (.xlsx) or CSV file. The platform loads the rows into a staging (import set) table, and a transform map then decides which spreadsheet column goes into which field of the target table, such as User or Incident.

The solution uses a data source to read the file, field maps to match the columns, a coalesce field to avoid duplicate records, and transform scripts to clean values such as dates before they are saved. After every run, the Transform History and import log show how many rows were inserted, updated, ignored or failed, so that the admin can fix the file and import again. The whole solution runs on a free ServiceNow Personal Developer Instance (PDI).

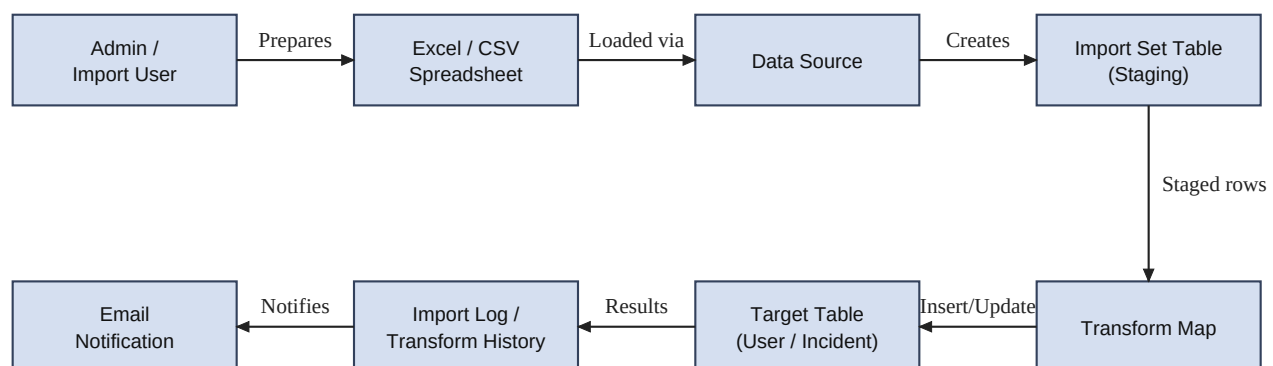
Scenarios

Scenario 1: The HR team sends an admin an Excel file with 500 new employee records. The admin uploads the file, creates a transform map to the User table, and runs it. All employee accounts are created in a few minutes without any manual typing.

Scenario 2: A service desk gets a weekly CSV file of tickets from a partner company. The transform map uses the ticket number as the coalesce field, so records that already exist are updated and only the new tickets are inserted, with no duplicates.

Scenario 3: A spreadsheet has dates in the dd/mm/yyyy format and a few rows with a blank email. A field script converts the dates, and an onBefore script skips the blank rows. The skipped rows are listed in the import log, so the admin can fix them and import again.

Technical Architecture:



Description: The solution follows a simple flow inside the ServiceNow platform. The admin prepares a spreadsheet and loads it through a data source, which creates a staging table called the import set table. A transform map reads the staged rows, applies the field maps and scripts, and inserts or updates the records in the target table. The results are stored in the import log and Transform History, and an email notification can be sent when the import is finished. Access to all these steps is controlled by ServiceNow roles and ACLs.

Pre-requisites:

- **ServiceNow Developer Account:** [ServiceNow Developer Site](#)
- **Import Sets and Transform Maps Knowledge:** [ServiceNow Product Documentation](#)
- **Basic JavaScript (for transform scripts):** [W3Schools JavaScript Tutorial](#)
- **Spreadsheet Skills:** [Microsoft Excel Support](#)
- **Web Browser:** Google Chrome (recommended)

Project Workflow:

Milestone 1: Environment Setup and Architecture

- **Activity 1.1:** Request a ServiceNow Personal Developer Instance (PDI) and log in as admin.
- **Activity 1.2:** Prepare the spreadsheet (clean headers, one value per cell, consistent date format).
- **Activity 1.3:** Define the architecture of the solution and the flow of the data.
- **Activity 1.4:** Plan the target table, the field mapping and the coalesce field.

Milestone 2: Data Source and Import Set

- **Activity 2.1:** Load the spreadsheet using Load Data to create the data source and import set table.
- **Activity 2.2:** Verify the staged rows in the import set table.

Milestone 3: Transform Map Development

- **Activity 3.1:** Create the transform map from the import set table to the target table.
- **Activity 3.2:** Create the field maps and set the coalesce field.
- **Activity 3.3:** Add transform scripts for value conversion and validation.

Milestone 4: Running and Validating the Import

- **Activity 4.1:** Run the transform on the import set.
- **Activity 4.2:** Review the Transform History, import log and target table.
- **Activity 4.3:** Fix failed rows and import them again.

Milestone 5: Deployment

- **Activity 5.1:** Capture the changes in an update set.
- **Activity 5.2:** Move the update set to another instance and test the import there.

Milestone 6: Conclusion

Milestone 1: Environment Setup and Architecture

In this milestone, we set up the ServiceNow instance that is used for the whole project, prepare the spreadsheet that will be imported, and decide how the data will move from the file to the target table.

Activity 1.1: Request a Personal Developer Instance

A Personal Developer Instance (PDI) is a free ServiceNow instance for learning and testing. Follow the steps below to get one.

- **Step 1 — Create a Developer Account:** Visit developer.servicenow.com and sign up for a free account.
- **Step 2 — Request an Instance:** After logging in, click on “Request Instance” and wait a few minutes until the instance is ready.
- **Step 3 — Log in as Admin:** Open the instance URL and log in with the admin username and password shown on the screen. Copy the password and keep it in a safe place.
- **Important** — A PDI is switched off or reclaimed if it is not used for a long time, so log in to it regularly while the project is going on.

Activity 1.2: Prepare the Spreadsheet

A clean spreadsheet makes the import much easier. Check the following before uploading the file:

- **Header row:** Keep the column names in the first row, with one header for each column and no merged cells.
- **Clean data:** Remove blank rows, extra sheets and totals at the bottom of the sheet.
- **Date format:** Use the same date format in the whole column (for example dd/mm/yyyy).
- **Unique key:** Keep a column with a unique value, such as the employee id, which is used later as the coalesce field.
- **File type:** Save the file as .xlsx or .csv.

An example of the sample data used in this project (employees.xlsx):

Emp ID	First Name	Last Name	Email	Department	Start Date
E1001	Anitha	Kumar	anitha.kumar@example.com	Finance	01/07/2026
E1002	Ravi	Shankar	ravi.shankar@example.com	IT	15/07/2026
E1003	Meena	Devi	meena.devi@example.com	HR	03/08/2026

Sample spreadsheet (example data)

Activity 1.3: Define the Architecture of the Solution

- **Draft the Data Flow:** Show how the file moves through the data source, import set table, transform map and target table.
- **Detail the Admin Steps:** Upload the file, create the transform map, run the transform and check the results.
- **Outline the Platform Responsibilities:** The import set table stores the raw rows, the transform map applies the rules, and the target table stores the final records.
- **Describe the Checks:** The coalesce field prevents duplicates, and the import log and Transform History record every skipped or failed row.

Activity 1.4: Plan the Target Table and Field Mapping

- **Choose the Target Table:** For this project, the target table is User [sys_user].
- **List the Mapping:** Emp ID to Employee number, First Name to First name, Last Name to Last name, Email to Email, Department to Department and Start Date to Start date.
- **Select the Coalesce Field:** Employee number is used, so that an existing user is updated instead of a new one being created.
- **Create an Update Set:** Before creating anything, create and select a new Local Update Set so that all the changes are captured for deployment in Milestone 5.

Milestone 2: Data Source and Import Set

In this milestone, the spreadsheet is loaded into ServiceNow. Loading the file creates a data source and an import set table, which holds the raw rows before they are transformed.

Activity 2.1: Load the Spreadsheet

- **Step 1 — Open Load Data:** In the application navigator, search for **System Import Sets** and open **Load Data**.
- **Step 2 — Create the Table:** Choose **Create table**, enter a label such as “Employee Import” and check the table name (for example u_employee_import).
- **Step 3 — Select the File:** Set the source of the import to **File**, the file format to **Excel (.xlsx)** (or CSV), and choose the file from your computer.
- **Step 4 — Set the Header Row:** Enter the sheet number and the header row number (1), then click **Submit**.
- **Step 5 — Wait for the Load:** ServiceNow loads the data and shows the import set. The rows are now stored in the staging table.

Activity 2.2: Verify the Staged Data

- Open the import set table list and check that the number of rows is the same as in the spreadsheet.

- Check that every column of the file became a field in the staging table and that the values look correct.
- If something is wrong, fix the spreadsheet and load it again before moving to the next milestone.

Milestone 3: Transform Map Development

Milestone 3 focuses on the transform map, which is the heart of this project. The transform map tells ServiceNow how to move each row from the import set table to the target table, and which rules to apply on the way.

Activity 3.1: Create the Transform Map

- **Step 1 — Open Transform Maps:** Go to **System Import Sets > Administration > Transform Maps** and click **New** (or use the Create Transform Map link on the loaded import set).
- **Step 2 — Fill the Details:** Enter the name (Employee Import Map), select the source table (u_employee_import) and the target table (User).
- **Step 3 — Set the Options:** Keep **Run business rules** and **Enforce mandatory fields** as required for the target table, then click **Submit**.

Activity 3.2: Create the Field Maps and Coalesce Field

- **Auto Map Matching Fields:** Use the related link to map the columns whose names match the target fields automatically.
- **Map the Remaining Fields:** Add field maps by hand for the columns that have different names, for example Emp ID to Employee number.
- **Set the Coalesce Field:** On the Employee number field map, select **Coalesce**. If a user with the same employee number already exists, it is updated; otherwise a new user is created.

Activity 3.3: Add Transform Scripts

Scripts are used when the value in the spreadsheet cannot be copied as it is. The first example is a field script that converts a dd/mm/yyyy date into the yyyy-mm-dd format:

```
answer = (function transformEntry(source) {  
    var d = source.u_start_date.toString();    // e.g. 15/07/2026  
    var p = d.split('/');  
    return p[2] + '-' + p[1] + '-' + p[0];      // 2026-07-15  
})(source);
```

The second example is an `onBefore` script. It skips a row if the email is empty and writes a message in the import log:

```
(function runTransformScript(source, map, log, target) {  
  if (source.u_email.nil()) {  
    ignore = true;  
    log.warn('Row skipped: email is empty');  
  }  
})(source, map, log, target);
```

Milestone 4: Running and Validating the Import

In this milestone, the transform map is run and the results are checked. This confirms that the data reached the target table correctly and shows which rows need to be fixed.

Activity 4.1: Run the Transform

- Open the import set that was created in Milestone 2.
- Click **Run Transform**, select the Employee Import Map and click **Transform**.
- Wait until the transform is completed. Big files may take a few minutes.

Activity 4.2: Review the Results

- **Transform History:** Open **System Import Sets > Transform History** to see how many rows were inserted, updated, ignored or failed.
- **Import Log:** Check the log messages for the reason behind every skipped or failed row.
- **Target Table:** Open the User list and confirm that the new records are there with the correct values.

Activity 4.3: Fix and Import Again

- Correct the rows that failed in the spreadsheet (for example, add the missing email or fix the date).
- Load the corrected file again and run the transform. Because the coalesce field is set, the records that were already imported are updated and not duplicated.

Milestone 5: Deployment

In Milestone 5, the transform map and the other changes are packaged and moved to another ServiceNow instance, for example from the PDI to a test or production instance. This is done with an update set.

Activity 5.1: Capture the Changes in an Update Set

- Open **System Update Sets > Local Update Sets** and check that the update set created in Milestone 1 contains the data source, the import set table and the transform map with its field maps and scripts.

- Change the state of the update set to **Complete** and use **Export to XML** to download it.

Activity 5.2: Move to Another Instance and Test

- On the other instance, go to **System Update Sets > Retrieved Update Sets** and choose **Import Update Set from XML**.
- Upload the XML file, open the update set, and click **Preview Update Set**. Solve the problems shown, if any.
- Click **Commit Update Set** and then run the import with the sample spreadsheet to confirm that everything works.

Important Notes:

- Always test with a small file (10 to 20 rows) before importing the full spreadsheet.
- The values in the coalesce field must be unique in the spreadsheet, otherwise the same record is updated again and again.
- Do not change the header row of the file after the transform map is created, as the field maps depend on the column names.
- The free PDI has limited resources, so split very large files into smaller ones.

Exploring the ServiceNow Screens:

The screens below are example illustrations of the main pages used in this project. They are drawn as examples and use sample data.

Load Data Screen:

Load Data

Import set table	Create table
Label	Employee Import
Name	u_employee_import
Source of the import	File
File format	Excel (.xlsx)
File	employees.xlsx
Sheet number	1
Header row	1
Submit	

Fig. 1 Load Data screen (example)

Description: This is the first screen of the import. The admin selects Create table, gives the table a label and name, chooses the file source, the file format and the spreadsheet file, and enters the sheet and header row numbers. On clicking Submit, the data is loaded into the import set table.

Transform Map and Field Maps:

Transform Map: Employee Import Map			
Source table:	u_employee_import	Target table:	User [sys_user]
Source field	Target field	Coalesce	Use source script
u_emp_id	Employee number	true	false
u_first_name	First name	false	false
u_last_name	Last name	false	false
u_email	Email	false	false
u_department	Department	false	false
u_start_date	Start date	false	true

Fig. 2 Transform map with field maps (example)

Description: The transform map connects the source table (import set) with the target table (User). Each row of the field map list shows one source field, the target field it goes to, whether it is the coalesce field, and whether a source script is used. In this example, Employee number is the coalesce field and Start date uses a script.

Transform History:

Transform History				
500	480	15	3	2
Total	Inserted	Updated	Ignored	Errors
Row	Status	Message		
42	Ignored	Row skipped: email is empty		
118	Ignored	Row skipped: email is empty		
260	Ignored	Row skipped: email is empty		
305	Error	Start date could not be converted		
391	Error	Start date could not be converted		

Fig. 3 Transform history and import log (example)

Description: After the transform runs, this page shows the summary of the result, with the total rows and the count of inserted, updated, ignored and error rows. The list below the summary gives the row number and the reason for each ignored or failed row, so that the admin can correct the spreadsheet and import again.

Conclusion:

Import Data Using Transform Maps is a simple and reliable way to move spreadsheet data into ServiceNow. By loading the file into an import set table and using a transform map with field maps, a coalesce field and transform scripts, the data reaches the target table correctly and without duplicates. The project, which included environment setup, data source creation, transform map development, validation and deployment with an update set, shows how much manual effort and how many data errors can be saved with a structured import process. In the future, the solution can be expanded with scheduled imports from a shared location, imports through web services, more validation rules, and dashboards that report on the import results.